

Md Sultanul Islam Ovi

PhD Applicant | AI/ML Researcher

 [portfolio](#) |  [scholar](#) |  [github](#) |  [linkedin](#) |  movi@gmu.edu |  (+1)5715919887 |  Virginia, USA

Research Interests

My research interests span applied AI, NLP, and human centered machine learning. Specifically, I work on: (a) AI/ML applications in education and healthcare; (b) AI systems for accessibility, wellbeing, and online safety; and (c) responsible and trustworthy AI for sensitive real world domains.

Education

George Mason University

M.S. in Computer Science (Current GPA: 4/4)

• Concentration: Machine Learning

Fairfax, VA, USA

Aug. 2023 – Present

Islamic University of Technology

B.Sc. in Computer Science and Engineering, Honors Class (CGPA: 3.87/4)

Gazipur, Bangladesh

Jan. 2017 – Mar. 2021

Publications

- Protecting Student Mental Health with a Context-Aware Machine Learning Framework for Stress Monitoring** 
Md Sultanul Islam Ovi, Jamal Hossain, Md Raihan Alam Rahi, Fatema Akter
2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking, IEEE, 2026
- Quantitative Currency Evaluation in Low-Resource Settings through Pattern Analysis to Assist Visually Impaired Users** 
Md Sultanul Islam Ovi, Mainul Hossain, Md Badsha Biswas
25th International Conference on Data Mining (ICDM), Workshop: Data Mining for Reliable Decision Making, IEEE, 2025
- Benchmarking ChatGPT, Codeium, and GitHub Copilot: A Comparative Study of AI-Driven Programming and Debugging Assistants** 
Md Sultanul Islam Ovi, Nafisa Anjum, Tasmina Haque Bithe, Md. Mahabubur Rahman, Mst. Shahnaj Akter Smrity
27th International Conference on Computer and Information Technology (ICCIT), IEEE, 2024
- PhishGuard: A Multi-Layered Ensemble Model for Optimal Phishing Website Detection** 
Md Sultanul Islam Ovi, Md. Hasibur Rahman, Mohammad Arif Hossain
6th International Conference on Sustainable Technologies for Industry 5.0 (STI), IEEE, 2024
- Sedimentation Analysis on Padma River Using Machine Learning Techniques** 
Md. Ruhen Hossain Bhuiyan, Niloy Saha Roy, Monoshi Kumar Roy, Md. Sultanul Islam Ovi, Faiz Al Faisal, Muhammad Aminur Rahaman
4th International Conference on Sustainable Technologies for Industry 4.0 (STI), IEEE, 2022

Submitted Papers

- A Dual Pipeline Machine Learning Framework for Automated Multi Class Sleep Disorder Screening Using Hybrid Resampling and Ensemble Learning** 
Md Sultanul Islam Ovi, Muhsina Tarannum Munfa, Miftahul Alam Adib, Syed Sabbir Hasan
Submitted to Journal (Under Review)
- Tiny LLMs for Efficient Low-Resource Bangla Cyberbullying and Harassment Analysis**
Md Sultanul Islam Ovi, Saif Siddique, Mahbuba Habib, Miraj Uddin Chowdhury, Shamazda Islam
Submitted to Journal (Under Review)
- Comparative Analysis of Tiny LLMs for Bangla Hate Speech and Emotion Detection Across Multiple Datasets**

Research Artifacts

1. **SkySeaLand Satellite Object Detection Dataset**, (DOI: 10.17632/d42n3cp86p.3) [[Mendeley Data](#)] [[Kaggle](#)]
Md. Zahid Hasan Riad, Md Sultanul Islam Ovi
 - Curated a high-resolution dataset of 1,300+ satellite images with annotations across four distinct classes.
 - Formatted annotations in YOLO and COCO standards using CVAT and Roboflow to facilitate benchmarking in mixed terrestrial and maritime domains.

Research Experience

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| George Mason University
<i>Graduate Research Assistant, Advisor: Prof. Sanjeev Setia</i> <ul style="list-style-type: none">• Focused on GPU sharing, distributed training strategies, and resource optimization for deep learning. | Fairfax, VA, USA
May 2024 – Aug. 2024 |
| Green University of Bangladesh
<i>Research Assistant, Systems & Security Research Cell, Advisor: Prof. Saiful Azad</i> <ul style="list-style-type: none">• Focused on system security, vulnerability analysis, and secure software practices. | Dhaka, Bangladesh
Jun. 2022 – Jun. 2023 |
| Islamic University of Technology
<i>Undergraduate Researcher, Advisor: Prof. Muhammad Mahbub Alam</i> <ul style="list-style-type: none">• Focused on network security, SDN architecture, and real-time attack detection. | Gazipur, Bangladesh
Jan. 2020 – Mar. 2021 |

Teaching Experience

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| George Mason University
<i>Graduate Teaching Assistant, Department of Computer Science</i> <ul style="list-style-type: none">• Facilitated lab sessions, conducted office hours, and graded assignments in Systems Programming and Object-Oriented Programming. | Fairfax, VA, USA
Aug. 2023 – Present |
| Green University of Bangladesh
<i>Lecturer, Department of Computer Science & Engineering</i> <ul style="list-style-type: none">• Courses Taught: Structured Programming, Data Structures, Algorithms, Computer Fundamentals & IT, Data Communications, Computer Networking, Discrete Mathematics, Digital Logic Design, Microprocessors & Micro-controllers, Database Systems. | Dhaka, Bangladesh
Jun. 2021 – Jun. 2023 |

Professional Experience

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| CETL - Green University
<i>Teaching & Learning Development Trainee</i> <ul style="list-style-type: none">• Trained in modern teaching strategies: Pedagogy, Andragogy, and Heutagogy.• Participated in faculty skill development and capacity-building workshops.• Engaged in collaborative teaching enhancement activities. | Dhaka, Bangladesh
Jun. 2021 – Sep. 2021 |
| Green University of Bangladesh
<i>Mentor Moderator</i> <ul style="list-style-type: none">• Monitored the mentoring activity of the mentors & developed mentorship guidelines.• Organized 'Meet the Mentors' program every semester.• Facilitated regular meetings with the Mentors, Director of Permanent Campus, and respective Chairpersons. | Dhaka, Bangladesh
Oct. 2021 – Jun. 2023 |
| Beetles Cyber Security Ltd.
<i>Cyber Security Intern, Supervisor: Mr. Shahee Mirza</i> <ul style="list-style-type: none">• Applied different strategies in Cyber Kill Chain Model and MITRE ATT&CK Model to defend systems under attack.• Performed vulnerability analysis on different compromised systems using open source tools like Nmap, Metasploit, Wireshark, Nessus etc. | Dhaka, Bangladesh
Nov. 2019 – Dec. 2019 |

Technical Skills

- **Programming Languages:** Python, Java, Golang, C/C++, SQL, Bash
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn
- **LLMs & Advanced AI:** LangChain, Agentic AI frameworks, RAG, Model Compression (Quantization, Pruning)
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI
- **Tools & Platforms:** AWS, GCP, Azure, Linux, Docker, Kubernetes, Git, MLflow
- **Distributed Computing:** Multi-GPU Training, HPC, MPI, Slurm, Ray, DeepSpeed

Technical Projects

NaturalQuery: AI-Powered SQL Query Generator 🔒

GPT-4, FastAPI, MySQL, Streamlit

- Built an AI system converting natural language to optimized SQL using GPT-4 with schema detection and validation.
- Implemented FastAPI backend with multi-database support and performance optimization suggestions.
- Deployed Streamlit web interface with real-time query execution, results visualization, and REST API endpoints.

PDF2Table: AI-Powered Data Extraction from Research Papers 🔒

RAG, GPT-4, ChromaDB, Streamlit

- Built end-to-end RAG pipeline extracting structured data (title, authors, summary, year) from research papers.
- Implemented vector search with ChromaDB and GPT-4o for intelligent document processing.
- Deployed production-ready Streamlit application with Docker containerization.

Math2LaTeX: Equation OCR to LaTeX with Qwen2 VL 🔒

PyTorch, Qwen2-VL, LoRA, Gradio

- Fine-tuned Qwen2-VL-7B vision-language model to convert mathematical equation images into LaTeX code.
- Developed interactive Gradio web app with LaTeX preview, clipboard integration, and .tex file export functionality.
- Leveraged LoRA adapters and 4-bit quantization for efficient training, reducing computational costs while maintaining performance.

Agentic RAG with CrewAI: Multi-Agent Knowledge Retrieval and Deployment 🔒

RAG, Groq, Tavily, AWS

- Built an agentic RAG system with CrewAI, routing between a ChromaDB knowledge base and Tavily web search.
- Developed a Streamlit interface and deployed the application on AWS EC2 using Docker for scalable access.
- Implemented collaborating agents for routing, retrieval, relevance grading, and hallucination checks using Groq LLaMA models.

Paperback Portal: E-commerce Bookstore Web Application 🔒

Java, JSP, Hibernate, MySQL

- Built a feature-rich e-commerce bookstore using Java EE, Hibernate ORM, and deployed on Tomcat Server.
- Added admin panel, role-based access, responsive UI, and secure payment integration; validated with JUnit.

GhostType: Cross-Platform Keylogger with Stealth Capabilities 🔒

Python, Cryptography

- Developed a stealth keylogger with RSA-encrypted logs, customizable input tracking, and auto self-destruct.
- Integrated secure email-based data delivery with user-defined intervals for forensic-grade usage.

LearningTurtles: School Management System 🔒

Node.js, Express.js, MongoDB

- Engineered a full-stack school management system with RESTful APIs for exams, registration, and tracking.
- Implemented role-based access and optimized MongoDB queries with pagination and filtering features.

Remaining projects are available on my [github](#) 🔒

Academic Services

Journals

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| • Reviewer, Asian Journal of Advanced Research and Reports | 2026 |
| • Reviewer, Health Informatics | 2026 |
| • Reviewer, Informatics for Health and Social Care | 2026 |
| • Reviewer, Mathematics and Computer Science | 2026 |

Conferences

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| • Reviewer, AIHH, UK | 2026 |
| • Program Committee Member, PEARC, USA | 2026 |
| • Reviewer, MICCAI, UAE | 2026 |

• <i>Reviewer</i> , IEC2T, India	2026
• <i>Reviewer</i> , AICTC, India	2026
• <i>Reviewer</i> , ICECET, Italy	2026
• <i>Reviewer</i> , ETNCC, Namibia	2026
• <i>Reviewer</i> , IEEE SoutheastCon, USA	2026
• <i>Reviewer</i> , Cumulus Athens, Greece	2026
• <i>Reviewer</i> , IEEE EDUCON, Egypt	2026
• <i>Reviewer</i> , IEEE i5CPS, India	2026
• <i>Reviewer</i> , IEEE IATMSI, India	2026
• <i>Reviewer</i> , IEEE BHI, USA	2025
• <i>Reviewer</i> , EAI BDCC, India	2025
• <i>Reviewer</i> , IEEE iConSCEPT, India	2025

Workshops

• <i>Reviewer</i> , ICML, AIWILD Workshop, South Korea	2026
• <i>Reviewer</i> , CVPR, MMRAgI Workshop, USA	2026
• <i>Program Committee Member</i> , CVPR, ADFM Workshop, USA	2026
• <i>Reviewer</i> , ICLR, ES-Reasoning Workshop, Brazil	2026
• <i>Reviewer</i> , ICLR, LLM Reasoning Workshop, Brazil	2026
• <i>Reviewer</i> , AAAI, MURE Workshop, Singapore	2026
• <i>Reviewer</i> , BMVC, MPI Workshop, UK	2025
• <i>Reviewer</i> , NeurIPS, LLM Evaluation Workshop, USA	2025
• <i>Reviewer</i> , NeurIPS, Muslims in ML Workshop, USA	2025
• <i>Reviewer</i> , ACL IJCNLP, BLP Workshop, India	2025

Professional Memberships

• <i>Member</i> , AAAI	Jun. 2025 – Present
• <i>Member</i> , IEEE	Apr. 2025 – Present
• <i>Member</i> , IEEE Computer Society	Apr. 2025 – Present
• <i>Member</i> , USENIX Association	Apr. 2025 – Apr. 2026

Volunteer & Outreach

AIML Professional Community Bangladesh 	Bangladesh (Remote)
<i>Community Mentor</i>	Mar. 2025 – Present
• Guiding AI/ML enthusiasts in research, career growth, and community building across Bangladesh's tech ecosystem.	

Certificates

• Natural Language Processing Specialization 	DeepLearning.AI
• DeepLearning.AI TensorFlow Developer 	DeepLearning.AI
• Mathematics for Machine Learning and Data Science Specialization 	DeepLearning.AI
• Python for Everybody Specialization 	University of Michigan
• Programming with Google Go Specialization 	University of California, Irvine
• Machine Learning A-Z: AI, Python & R 	Udemy

Extracurricular Activities

• <i>Member Secretary</i> , Hostel Infrastructure and Management Committee at GUB	2021 – 2023
• <i>Member</i> , Community Outreach Committee at GUB	2022 – 2023
• <i>Member</i> , Lab Development Committee at GUB	2022 – 2023
• <i>Technical Committee Member</i> , IEEE STI Conference, Bangladesh	2021 – 2022
• <i>Organizing Committee Member</i> , 9th and 10th National ICT Fest at IUT	2017 – 2019
• <i>Organizing Committee Member</i> , App Development Workshop at IUT	2019
• <i>4th Place</i> , National BDAPPS Innovation Hackathon	2018